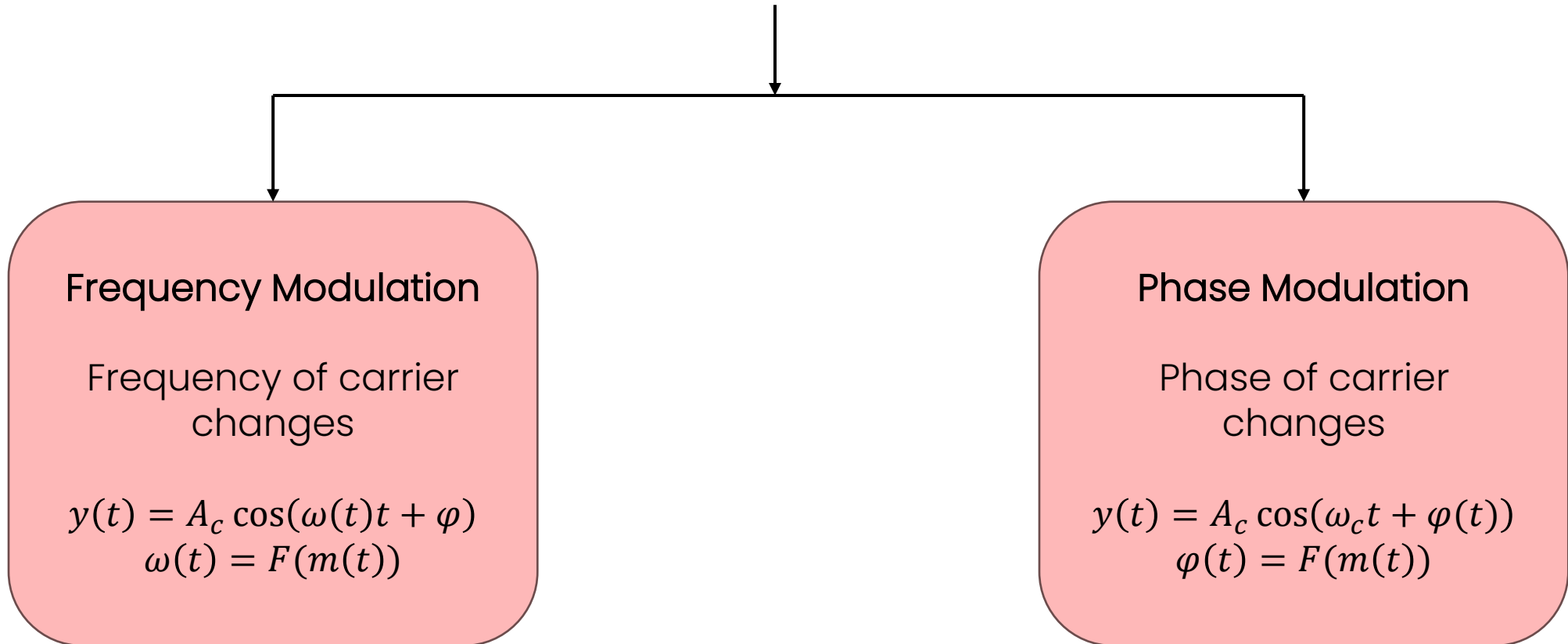
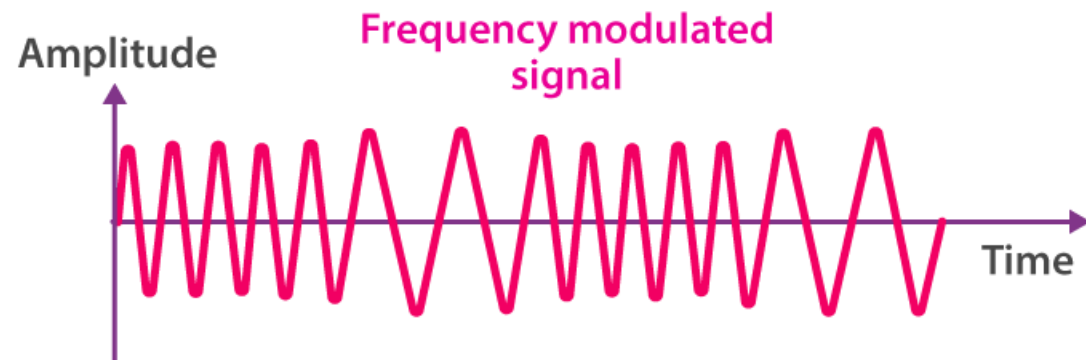
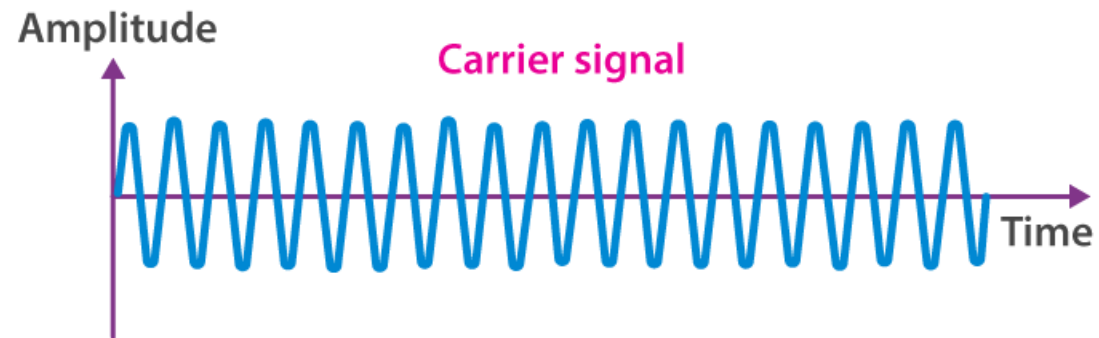
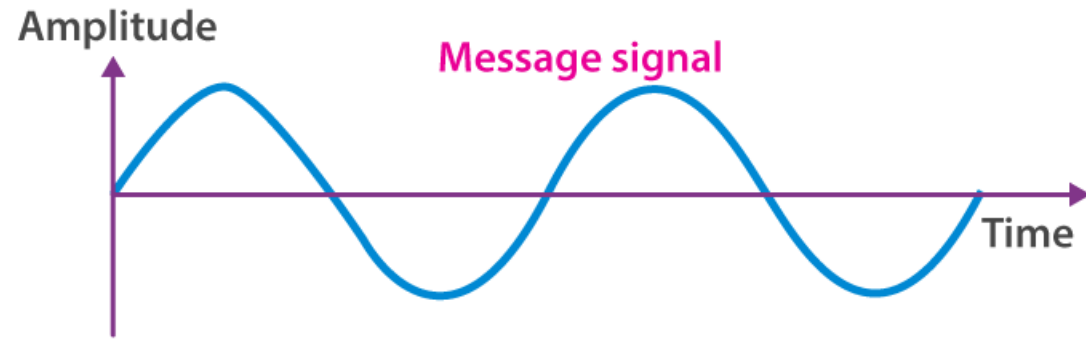


ANGLE MODULATION

Angle Modulation is a modulation process where the frequency or phase of the carrier signal changes with respect to message signal.



FREQUENCY MODULATION



FREQUENCY MODULATION

Advantages:

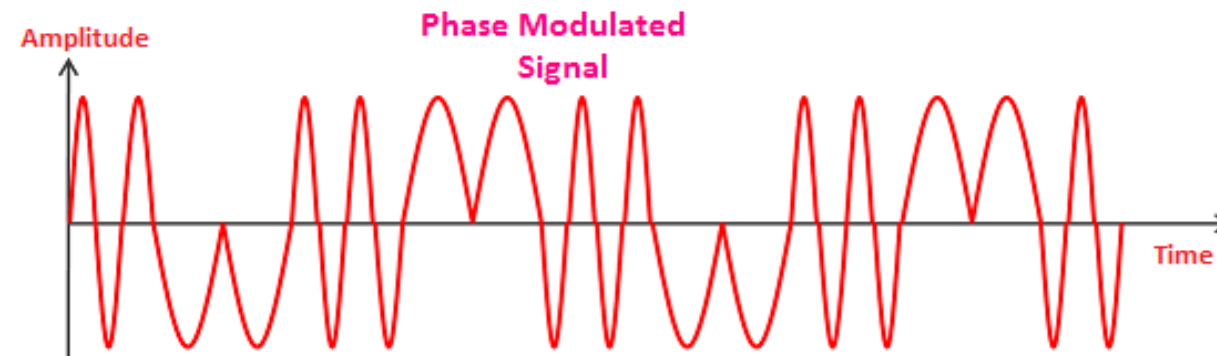
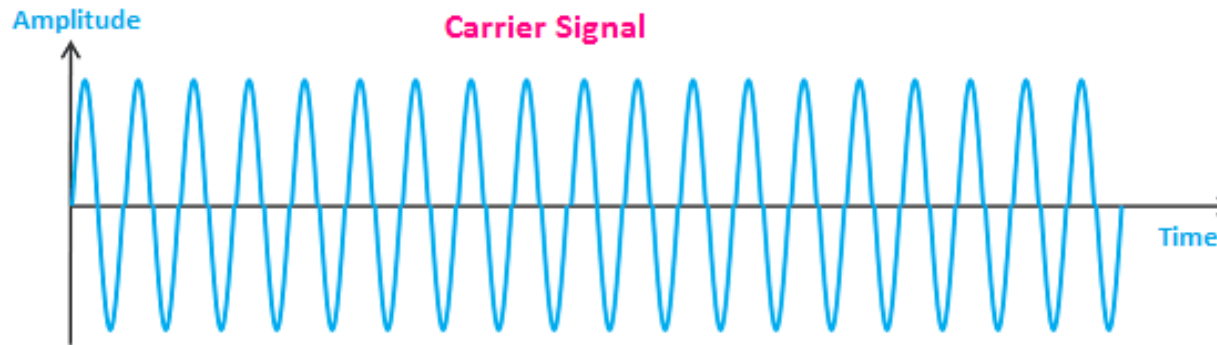
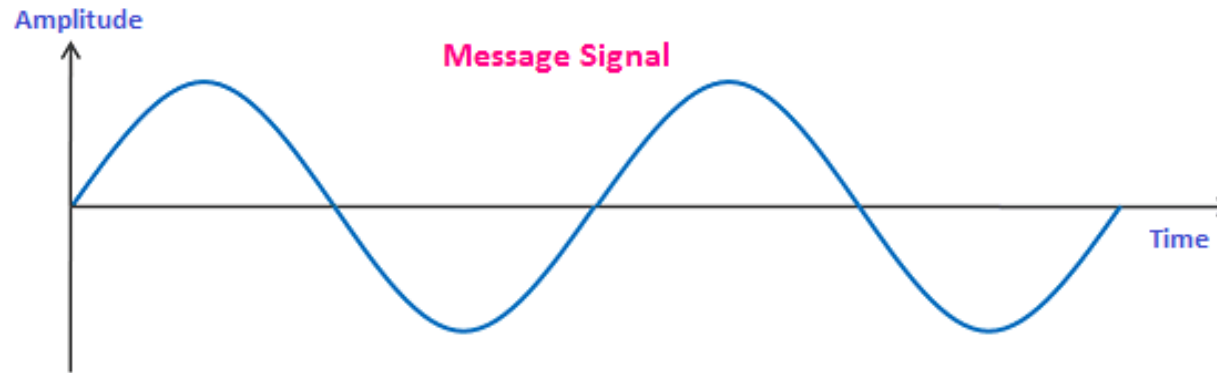
- Amplitude remains constant hence better noise immunity.
- More efficient use of bandwidth.
- Preferred for radio broadcasting due to superior quality and capacity than AM.
- Less power consumption than AM.

Disadvantages:

- FM works with higher bandwidth which makes it costlier.
- Antennas need to be closer for clear communication.
- Complex transmitter-receiver settings.

PHASE MODULATION

Phase Modulation



PHASE MODULATION

Advantages:

- Fast operation.
- Circuit is less complicated compared to FM.
- Higher efficiency than AM.

Disadvantages:

- PM is less noise resistant than FM.
- Poor signal to noise ratio.
- Synchronization needed.
- Not widely used.

COMPARISON OF DIFFERENT METHODS

Feature	Amplitude Modulation	Frequency Modulation	Phase Modulation
Modulated parameter	Amplitude	Frequency	Phase
Noise immunity	Poor	Excellent	Good
Efficiency	Low	High	High
Complexity	Simple	High	High
Demodulation	Envelope or coherent	Phase locked loop	Zero crossing detector



THANK YOU

